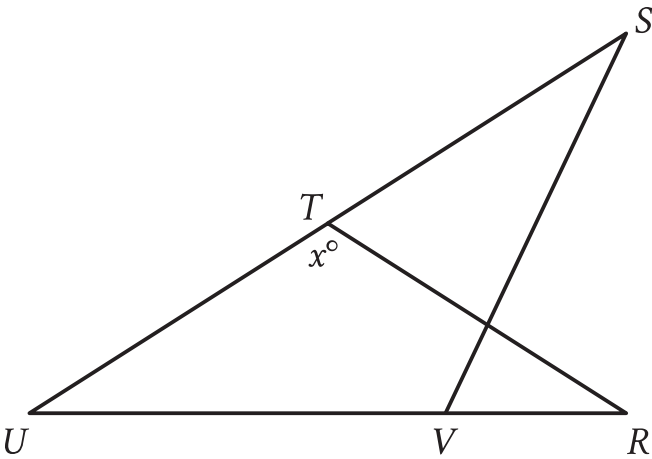


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $RT = TU$, the measure of angle VST is 29° , and the measure of angle RVS is 41° . What is the value of x ?

Correct Answer: 156

Rationale

The correct answer is **156**. In the figure shown, the sum of the measures of angle UVS and angle RVS is 180° . It's given that the measure of angle RVS is 41° . Therefore, the measure of angle UVS is $(180 - 41)^\circ$, or 139° . The sum of the measures of the interior angles of a triangle is 180° . In triangle UVS , the measure of angle UVS is 139° and it's given that the measure of angle VST is 29° . Thus, the measure of angle VUS is $(180 - 139 - 29)^\circ$, or 12° . It's given that $RT = TU$. Therefore, triangle TUR is an isosceles triangle and the measure of VUS is equal to the measure of angle TRU . In triangle TUR , the measure of angle VUS is 12° and the measure of angle TRU is 12° . Thus, the measure of angle UTR is $(180 - 12 - 12)^\circ$, or 156° . The figure shows that the measure of angle UTR is x° , so the value of x is **156**.